

In a nutshell: Implementing smart manufacturing

- Optimise existing processes before installing smart equipment
- Identify one pilot area first to do the implementation instead of starting with heavy investment
- Prepare a roadmap for present and future systems. Invest in machines that are upgradable rather than going for 'jugaad'/shortcuts
- Measure benefit/ROI in terms of improvement in productivity, customer satisfaction, and market share
- If you see value in the results, management can train the employees and gradually deploy smart manufacturing to other areas as well

mation to incorporate minute product variations without having to reassemble equipment provides flexibility in the production process. The manufacturers can thus produce customised products at reduced downtime and better efficiency. Manufacturing processes will be better; for instance, failure detection using automated optical inspection (AOI) can be done based on images. The main benefit of smart manufacturing is that investment, if done right, can decrease manufacturing costs significantly.

Today's quality-conscious customers want to know everything, from design to production. They develop a sense of confidence from the awareness that they are purchasing good quality products. Shrikant Sharadrao Borkar, founder, SSB Technocrat, says, "Customers want transparency in the process. With smart equipment, data can be shared in real time. The customer will be happy to interact, and for any changes, he/she can see the process remotely. This will encourage customers to interact at your end."

Talking about the benefits for manufacturers, Nikhil Pal says, "When we are using smart manufacturing initiatives for digital transformation, we are able to link the vendors in the digital supply chain. We get an overview of the vendor to the end customer. End-to-end visibility in the entire supply-value chain because of digitalisation allows us to see even the inventory level so that the focus is to better the efficiency, productivity and also minimise the inventory holding in our factories."

Overall, this is a win-win situation for both manufacturers and customers. When customers know that the manufacturer is using smart manufacturing technology/equipment for manufacturing, it increases their trust

and satisfaction, which enhances the brand value for the business in the long run. When machines are better in terms of speed, size, and capability, even workers will benefit.

Challenges that need to be dealt with

Companies realising the benefits of smart manufacturing and wanting to improve their work efficiency often find themselves stuck in the early stages of adoption. **MSMEs are the backbone of an economy, but the implementation of smart manufacturing in India is slower because of a lack of motivation in MSMEs.**

Shrikant Sharadrao Borkar says, "Adoption is low for SMEs due to inadequate infrastructure, use of refurbished machinery, and inability to justify ROI. However, the adoption rate might increase due to customer demand and globalisation requirements."

Due to the lack of adequate budget, companies tend to go for options that may not be fail-safe for the entire system, from design to production in the long run. The inability to justify ROI arises from the fact that traditionally companies follow the wrong approach of measuring automation benefits against a reduction in workforce.

Besides, there is a huge skill gap that acts as an impediment to the growth of the manufacturing sector, according to Industry Pulse: 2018 Manufacturing Workforce Report by Tooling U-SME, a provider of manufacturing training solutions.

Another challenge is intellectual property (IP) protection, that is, how are businesses going to keep proprietary data in smart factories. Roberto Gatti, CEO, Delvitech SA, says, "Customers need access to the systems like

the data collection system in the manufacturing plant to know the ongoing status of production. The amount of data coming in and going out will be huge, which requires proper infrastructure. At the same time, the privacy of individual customers is at risk if data in the manufacturing plant is not taken care of."

Current adoption scenario

There has been a big push for Industry 4.0 in the last ten years. Even though there is growing adoption of smart manufacturing worldwide, the market size varies considerably from one region to another.

Germany is projected to have the largest smart factory market share in the European region in the next five years based on a report from Global Market Insights. The report also states the current dominance of North America; the continent's smart factory market was valued at around thirteen billion dollars in 2018.

A significant portion of smart manufacturing companies is present in the Indo-Pacific region that includes countries like Japan, South Korea, Taiwan, and China. According to an IDC study, industrial companies across the region are undergoing a rapid digital transformation, adding 1.16 trillion dollars to Indo-Pacific GDP and boosting annual growth rates by 0.8 per cent. In Singapore, the government has come out with a smart industry readiness index (SIRI). They assess all industries based on this assessment. This is helping the government and companies in identifying how to start, scale, and sustain their transformation."

Yet, adoption in India is still low as compared to other countries, and manufacturers show a lack of interest in smart manufacturing. However, government initiatives like Smart Advanced Manufacturing and Rapid Transformation Hub (SAMARTH) Udyog Bharat 4.0 that aims to create awareness and integrate Industry 4.0 technologies in the Indian manufacturing sector for large organisations as well as MSMEs have a positive influence on the market. In the last three years since the operation, tax reform rolled out in the form of the Goods and Services Tax (GST) has had a huge impact on supply chains.